



Guangdong Technion

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GTIIT

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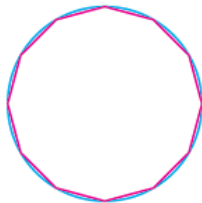
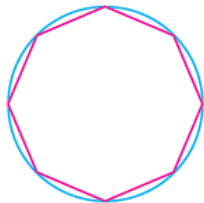
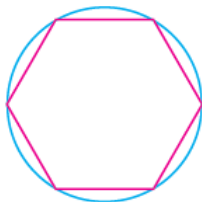
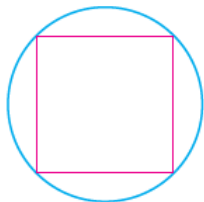


ARC LENGTH

Arc Length

- What do we mean by the **length of a curve**? We might think of fitting a piece of string to the curve and then measuring the string against a ruler. But that might be difficult to do with much accuracy if we have a complicated curve.
- We need a precise definition for the **length of an arc of a curve**, in the same spirit as the definitions we developed for the concepts of area and volume.
- If the curve is a **polygon**, we can easily find its length; we just add the lengths of the line segments that form the polygon: use the **distance formula to find the distance between the endpoints of each segment**.

Arc Length



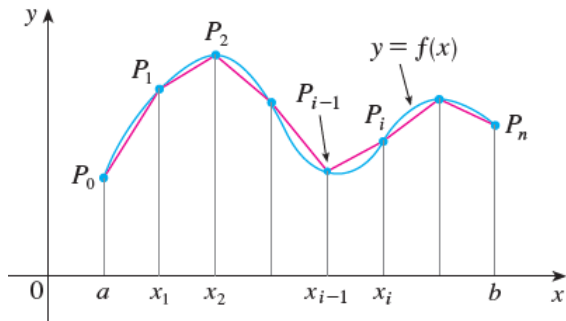
Arc Length

- We are going to find the length of a general curve by first approximating it by a polygon and then taking a limit as the number of segments of the polygon is increased.
- This process is familiar for the case of a circle, where the circumference is the limit of lengths of inscribed polygons.

Arc Length

- Now suppose that a curve C is defined by the equation $y = f(x)$, where f is continuous and $a < x < b$.
- We obtain a polygonal approximation to C by dividing the interval $[a, b]$ into n subintervals with endpoints $x_0, x_1, \dots, x_n$ and equal width Δx .
- If $y_k = f(x_k)$, then the point $P_k = (x_k, y_k)$ lies on C and the polygon with vertices $P_0, P_1, \dots, P_n$, is an approximation to C .
- The length L of C is approximately the length of this polygon and the approximation gets better as we let n increase.

Arc Length



Arc Length

We define the **length L of the curve C with equation $y = f(x)$** , where f is continuous and $a < x < b$, as the limit of the lengths of these inscribed polygons, provided the limit exists:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left\| P_k - P_{k-1} \right\| = \lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{(x_k - x_{k-1})^2 + (y_k - y_{k-1})^2} \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{(\Delta x)^2 + (\Delta y_k)^2} \end{aligned}$$

Arc Length

For computational purposes, but we can derive an integral formula for L in the case where f has a continuous derivative:

Such a function f is called **smooth** because a small change in x produces a small change in $f'(x)$

$$\text{Arc Length } L = \lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{(\Delta x)^2 + (\Delta y_k)^2}$$

By applying the **Mean Value Theorem** to f on the interval $[x_{k-1}, x_k]$, we find that there is a number x_k^* on $[x_{k-1}, x_k]$ such that

$$f(x_k) - f(x_{k-1}) = f'(x_k^*)(x_k - x_{k-1})$$

Then, we obtain

$$\begin{aligned} \Delta y_k &= f'(x_k^*)(\Delta x) \\ \implies \sqrt{(\Delta x)^2 + (\Delta y_k)^2} &= \sqrt{(\Delta x)^2 + [f'(x_k^*)(\Delta x)]^2} \\ &= \sqrt{(\Delta x)^2} \sqrt{1 + [f'(x_k^*)]^2} = \Delta x \sqrt{1 + [f'(x_k^*)]^2} \end{aligned}$$

Arc Length

If f' is continuous on $[a, b]$, then the length of the curve $y = f(x)$ on $[a, b]$ is

$$L = \lim_{n \rightarrow \infty} \sum_{k=1}^n \Delta x \sqrt{1 + [f'(x_k^*)]^2}$$

We recognize this as the limit of a Riemann Sum

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

In Leibniz notation for differentials:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

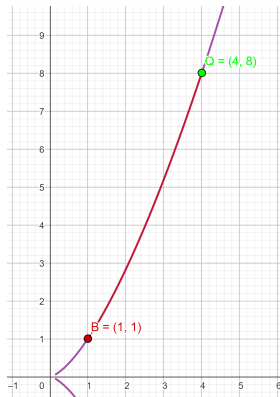
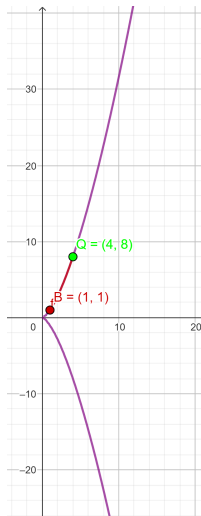
Arc Length

If a curve has the equation $x = g(y)$, $c < y < d$, and $g'(y)$ is continuous, then by interchanging the roles of x and y in the previous formula, we obtain the following formula for its length:

$$L = \int_c^d \sqrt{1 + [g'(y)]^2} dy$$

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Arc Length Ex 1. Find the length of the semicubical parabola $y^2 = x^3$ between the points $(1, 1)$ and $(4, 8)$.



Arc Length

For the top half of the curve, we can solve $y = f(x)$:

$$y^2 = x^3 \implies y = x^{3/2} \implies \frac{dy}{dx} = \frac{3}{2}x^{1/2} \implies \left(\frac{dy}{dx}\right)^2 = \frac{9}{4}x$$

And so, the arc length has the value

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

$$= \int_1^4 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_1^4 \sqrt{1 + \frac{9}{4}x} dx$$

Solve the integral by Substitution

- $u = 1 + \frac{9}{4}x \implies du = \frac{9}{4} dx$

- For the endpoints:

$$x = 1 \implies u = \frac{13}{4} \text{ and } x = 4 \implies u = 10$$

Therefore, the arc length is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_1^4 \sqrt{1 + \frac{9}{4}x} dx = \int_{\frac{13}{4}}^{10} \sqrt{u} \frac{4}{9} du$$

$$= \frac{4}{9} \frac{2}{3} u^{\frac{3}{2}} \Big|_{\frac{13}{4}}^{10} = \frac{8}{27} \left[10^{\frac{3}{2}} - \left(\frac{13}{4}\right)^{\frac{3}{2}} \right] = \frac{1}{27} \left[80\sqrt{10} - 13\sqrt{13} \right] \checkmark$$

ARC LENGTH

Ex 2. a) Solve $\int \sec^3 x \, dx$.

b) Find the length of the parabola $y^2 = x$ between the points $(0, 0)$ and $(1, 1)$.

a. Integrate by parts: $\begin{cases} u = \sec x \\ du = \sec(x) \tan(x) dx \end{cases}$ and $\begin{cases} dv = \sec^2 x \, dx \\ v = \tan x \end{cases}$

$$\Rightarrow \int \sec^3(x) dx = \sec x \tan x - \int \sec(x) \tan^2(x) dx$$

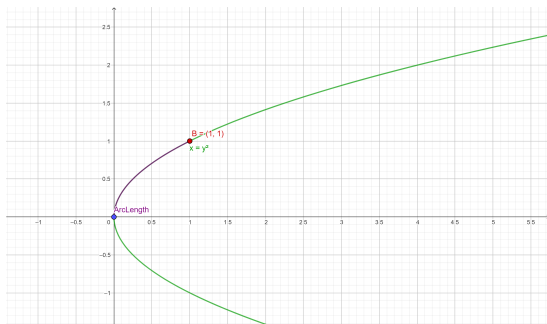
$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) dx$$

$$= \sec x \tan x - \int \sec^3 x \, dx + \underbrace{\int \sec x \, dx}_{= \ln |\sec x + \tan x|}$$

$$\Rightarrow \int \sec^3 x \, dx = \frac{1}{2} \left[\sec x \tan x + \ln |\sec x + \tan x| \right]$$

ARC LENGTH

b) Length of the parabola $y^2 = x$ between the points $(0, 0)$ and $(1, 1)$:



Use the formula for $x = g(y) = y^2$

$$\Rightarrow \frac{dx}{dy} = 2y \text{ and } \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \sqrt{1 + 4y^2}$$

Solve the integral by Trigonometric Substitution

And so, the arc length has the value

$$L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dx = \int_0^1 \sqrt{1 + 4y^2} dy$$

$$\bullet y = \frac{1}{2}\tan(t) \Rightarrow dy = \frac{1}{2}\sec^2(t)dt \Rightarrow \sqrt{1 + 4y^2} = \sqrt{1 + \tan^2(t)} = \sec(t)$$

• For the endpoints:

$$y = 0 \Rightarrow t = 0 \text{ and } 1 = y = \frac{1}{2}\tan(t) \Rightarrow \alpha = \arctan(2)$$

Therefore, the arc length is

$$L = \int_0^1 \sqrt{1 + 4y^2} dy = \int_0^\alpha \sec(t) \frac{1}{2} \sec^2(t) dt = \frac{1}{2} \int_0^\alpha \sec^3(t) dt$$

Arc Length: the last computation was made in a)

$$L = \int_0^1 \sqrt{1 + 4y^2} \, dy = \frac{1}{2} \int_0^\alpha \sec^3(t) \, dt$$

$$= \frac{1}{4} \left[\sec(t) \tan(t) + \ln \left| \sec(t) + \tan(t) \right| \right] \Big|_0^\alpha$$

$$= \frac{1}{4} \left[\sec(\alpha) \tan(\alpha) + \ln \left| \sec(\alpha) + \tan(\alpha) \right| \right]$$

Finally, since

$$\tan(\alpha) = 2 \implies \sec^2(\alpha) = 1 + \tan^2(\alpha) = 5 \implies \sec(\alpha) = \sqrt{5}$$

Hence,

$$\implies L = \frac{\sqrt{5}}{2} + \frac{\ln(\sqrt{5} + 2)}{4} \quad \checkmark$$

THE ARC LENGTH FUNCTION

- 1 Given f such that f' is continuous on $[a, b]$, the **Arc Length Function** measures the arc length of the curve $y = f(x)$ from a particular starting point $(a, f(a))$ to any other point $(x, f(x))$ on the curve, for $a < x < b$:

$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2} dt$$

- 2 By the Fundamental Theorem of Calculus:

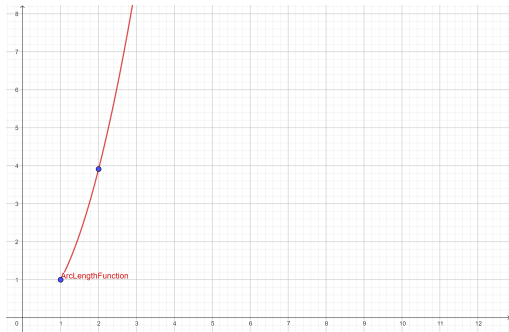
$$\frac{ds}{dx} = \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \text{ and } L = \int ds$$

- 3 If $x = g(y) \implies s(y) = \int_c^y \sqrt{1 + [g'(t)]^2} dt$ and

$$\frac{ds}{dy} = \sqrt{1 + [g'(y)]^2} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}$$

ARC LENGTH

Ex 3. a) Find the arc length function for the curve $y = x^2 - \frac{1}{8} \ln(x)$ starting at $P = (1, 1)$



ARC LENGTH

Arc length function of the curve $y = x^2 - \frac{1}{8} \ln(x)$ starting at $P = (1, 1)$

Solution. • $f(x) = x^2 - \frac{1}{8} \ln(x) \implies f'(x) = 2x - \frac{1}{8} \frac{1}{x}$, then

$$\bullet 1 + [f'(x)]^2 = 1 + \left(2x - \frac{1}{8x}\right)^2 \implies \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \left(2x - \frac{1}{8x}\right)^2}$$

$$= \sqrt{1 + 4x^2 - \frac{1}{2} + \frac{1}{64x^2}} = \sqrt{\left(2x + \frac{1}{8x}\right)^2} = \left(2x + \frac{1}{8x}\right)$$

• Hence, the arc length function is

$$s(x) = \int_1^x \sqrt{1 + [f'(t)]^2} dt = \int_1^x \left(2t + \frac{1}{8t}\right) dt = t^2 + \frac{1}{8} \ln(t) \Big|_1^x$$

$$= x^2 + \frac{1}{8} \ln(x) - 1$$

ARC LENGTH

Ex 3. b) Find the arc length function for the curve $y = x^2 - \frac{1}{8} \ln(x)$ from $(1, 1)$ to $(2, 4)$

Solution. • The arc length function **for any $x > 0$** is

$$s(x) = \int_1^x \sqrt{1 + [f'(t)]^2} dt = \int_1^x \left(2t + \frac{1}{8t}\right) dt = t^2 + \frac{1}{8} \ln(t) \Big|_1^x$$

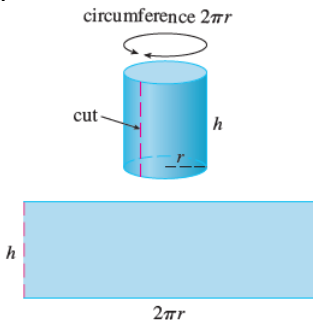
$$= x^2 + \frac{1}{8} \ln(x) - 1$$

• The arc length function **from $(1, 1)$ to $(2, 4)$** is then

$$s(2) = 2^2 + \frac{1}{8} \ln(2) - 1 = 3 + \frac{\ln(2)}{8}$$

AREA OF A SURFACE OF REVOLUTION

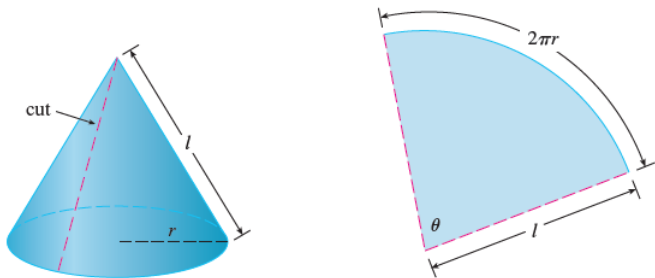
- A **surface of revolution** A is obtained by the rotation of a curve around a line.
- Such a surface is the lateral boundary of a **solid of revolution**.
- We realize that covering the surface would require the **same amount of paper as does a flat region with area A** .
- The lateral surface area of a circular cylinder with **radius r** and **height h** is $A = 2\pi rh$:
cut the cylinder and unrolling it to obtain a **rectangle** with dimensions $2\pi r$ and h .



AREA OF A SURFACE OF REVOLUTION

- For a circular cone with base radius r and slant height ℓ , cut it along a line, and flatten it to form a **sector of a circle with radius r and central angle $\theta = 2\pi r/\ell$** , for which the area is

$$A = \frac{1}{2}\ell^2\theta = \frac{1}{2}\ell^2\left(\frac{2\pi r}{\ell}\right) = \pi r\ell$$



AREA OF A SURFACE OF REVOLUTION

- For more complicated surfaces of revolution: follow the strategy of approximating the original curve by a polygon.
- When this polygon is rotated about an axis, it creates a simpler surface whose area approximates the actual area.
- The limit, as the amount of vertices $n \rightarrow \infty$, determine the **exact surface area**.
- The approximating surface consists of a number of bands, each formed by **rotating a line segment about an axis**.
- To find the surface area, each of these bands can be considered a portion of a circular cone.

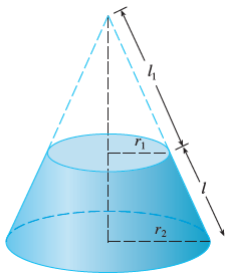
AREA OF A SURFACE OF REVOLUTION

- The **area of the band**, a frustum of a cone, with **slant height** ℓ and **upper and lower radii** r_1 and r_2 is found by subtracting the areas of two cones:

$$A = \pi r_2(\ell_1 + \ell) - \pi r_1 \ell_1 = \pi \left[(r_2 - r_1)\ell_1 + r_2 \ell \right]$$

- For similar triangles: $\frac{\ell_1}{r_1} = \frac{\ell_1 + \ell}{r_2} \implies (r_2 - r_1)\ell_1 = r_1 \ell$
- Then, choosing $r = \frac{r_1 + r_2}{2}$, the **average radius of the band**:

$$A = \pi(r_1 \ell + r_2 \ell) \implies A = 2\pi r \ell$$

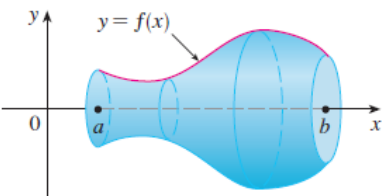


DEFINITION OF AREA OF A SURFACE OF REVOLUTION

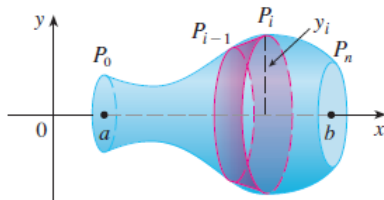
- Now we apply this formula to our strategy: Consider the surface obtained by rotating the curve $y = f(x)$ on $[a, b]$, around the x -axis, where $f > 0$ and has a continuous derivative f' .
- Divide $[a, b]$ into n subintervals with endpoints $x_0, x_1, \dots, x_n$ and equal width Δx .
- Denote by $P_i = (x_i, y_i) = (x_i, f(x_i))$ the $n + 1$ points on the curve.
- The part of the surface between x_i and x_{i+1} is approximated by rotating the line segment from P_i to P_{i+1} around the x -axis.
- The result is a band with slant height $\ell = ||P_{i+1} - P_i||$ and average radius $r = \frac{1}{2}(y_i + y_{i+1})$ with surface area:

$$A_i = 2\pi r \ell = 2\pi \frac{1}{2}(y_i + y_{i+1}) ||P_{i+1} - P_i||$$

AREA OF A SURFACE OF REVOLUTION



Surface of Revolution



Approximating Band

AREA OF A SURFACE OF REVOLUTION

- Each approximating band of slant height $\ell = \|P_{i+1} - P_i\|$ and average radius $r = \frac{1}{2}(y_i + y_{i+1})$ has surface area:

$$A_i = 2\pi r \ell = 2\pi \frac{1}{2}(y_i + y_{i+1}) \|P_{i+1} - P_i\|$$

- As we have computed for the arc length, as $n \rightarrow \infty$ and Δx is small:

$$\frac{1}{2}(y_i + y_{i+1}) \approx f(x_i^*) \quad \text{and} \quad \|P_{i+1} - P_i\| = \sqrt{1 + [f'(x_i^*)]^2} \Delta x$$

Then, the approximating band has surface area:

$$A_i = 2\pi \underbrace{\frac{1}{2}(y_i + y_{i+1})}_{=f(x_i^*)} \underbrace{\|P_{i+1} - P_i\|}_{=\sqrt{1+[f'(x_i^*)]^2}\Delta x}$$

AREA OF A SURFACE OF REVOLUTION

- The approximating band has area surface:

$$A_i = 2\pi \underbrace{\frac{1}{2}(y_i + y_{i+1})}_{=f(x_i^*)} \underbrace{||P_{i+1} - P_i||}_{=\sqrt{1+[f'(x_i^*)]^2} \Delta x} = 2\pi f(x_i^*) \sqrt{1 + [f'(x_i^*)]^2} \Delta x$$

- An approximation of the area of the complete surface of revolution is then the sum of all these small areas:

$$S \approx \sum_{i=1}^n 2\pi f(x_i^*) \sqrt{1 + [f'(x_i^*)]^2} \Delta x$$

- This approximation becomes better as $n \rightarrow \infty$ and, recognizing the sum as a Riemann sum, we obtain:

$$S = \lim_{n \rightarrow \infty} \sum_{i=1}^n 2\pi f(x_i^*) \sqrt{1 + [f'(x_i^*)]^2} \Delta x = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx \quad \checkmark$$

AREA OF A SURFACE OF REVOLUTION

- Let $f > 0$ be a differentiable function such that f' is continuous on $[a, b]$ and consider the revolution surface obtained by the rotation of the curve $y = f(x)$ around the x -axis.

The area of the revolution surface is by definition given by

$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

- If the curve is described in the form $x = g(y)$, where g is differentiable, g' is continuous on $[c, d]$ and the rotation is about the y -axis:

$$S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

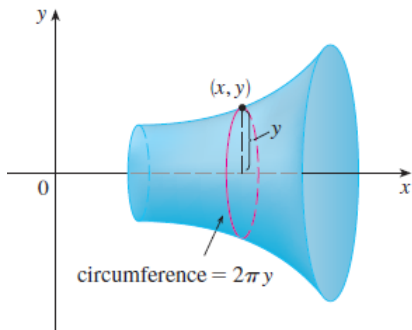
AREA OF A SURFACE OF REVOLUTION

Recall $\frac{ds}{dx} = \sqrt{1 + [f'(y)]^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

$$\Rightarrow ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

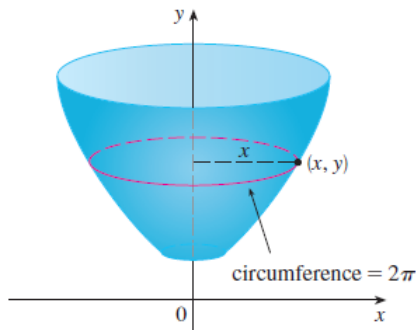
$$\Rightarrow S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b 2\pi y ds \checkmark$$

AREA OF A SURFACE OF REVOLUTION



(a) Rotation about x-axis: $S = \int 2\pi y \, ds$

$$S = \int_a^b 2\pi y \, ds$$



(b) Rotation about y-axis: $S = \int 2\pi x \, ds$

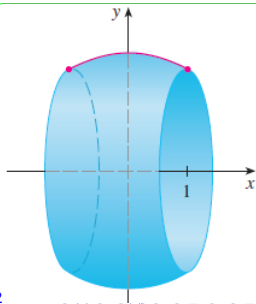
$$S = \int_c^d 2\pi x \, ds$$

AREA OF A SURFACE OF REVOLUTION

Ex 4. Find the area of the surface obtained by rotating the arc of the circle $x^2 + y^2 = 4$, $-1 \leq x \leq 1$ around the x -axis.

Solution • $y = f(x) = \sqrt{4 - x^2} \implies f'(x) = \frac{dy}{dx} = \frac{-2x}{2\sqrt{4-x^2}}$

$$\implies S = \int_{-1}^1 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2\pi \int_{-1}^1 \sqrt{4 - x^2} \sqrt{1 + \frac{x^2}{4 - x^2}} dx$$



The surface is a portion of the sphere of radius 2

AREA OF A SURFACE OF REVOLUTION

$$\Rightarrow S = 2\pi \int_{-1}^1 \sqrt{4-x^2} \sqrt{1 + \frac{x^2}{4-x^2}} dx$$

$$= 2\pi \int_{-1}^1 \sqrt{4-x^2} \sqrt{\frac{4-x^2+x^2}{4-x^2}} dx = 2\pi \int_{-1}^1 \sqrt{4-x^2+x^2} dx$$

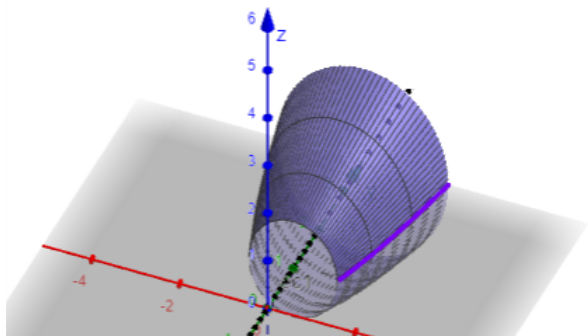
$$= 2\pi \int_{-1}^1 \sqrt{4} dx = 4\pi \int_{-1}^1 1 dx = 8\pi \checkmark$$

AREA OF A SURFACE OF REVOLUTION

Ex 5. Find the area of the surface generated by the rotation of the arc of the parabola $y = x^2$, from $(1, 1)$ to $(2, 4)$, around the y -axis.

Solution. • The interval for y will be $[1, 4]$

• $x = g(y) = \sqrt{y} \implies g'(y) = \frac{dx}{dy} = \frac{1}{2\sqrt{y}}$



AREA OF A SURFACE OF REVOLUTION

$$\Rightarrow S = \int_1^4 2\pi g(y) \sqrt{1 + [g'(y)]^2} dy = \int_1^4 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$= 2\pi \int_1^4 \sqrt{y} \sqrt{1 + \frac{1}{4y}} dy = 2\pi \int_1^4 \frac{1}{2} \sqrt{4y + 1} dy = [*]$$

- Substitution $u = 1 + 4y \Rightarrow du = 4 dy$, $u(1) = 5$, $u(4) = 17$

$$\Rightarrow [*] = \frac{\pi}{4} \int_5^{17} \sqrt{u} du = \frac{\pi}{6} u^{3/2} \Big|_5^{17} = \frac{\pi}{6} (17\sqrt{17} - 5\sqrt{15}) \checkmark$$